

Mayhem Attribute Conflation

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What We Started With

- Goal: choose the better value between current and base for name, phone, website, address, and category
- Starting system had:
 - rule-based baselines: Most Recent, Completeness, Confidence, heuristic Hybrid
 - per-attribute ML trained on synthetic data
- Strongest starting performance came from simple heuristics, not ML
- I did not want production logic to rely mainly on recency, even if it scored well offline

Attribute	Most Recent F1	Heuristic Hybrid F1	Starting ML F1
Address	0.8338	0.8338	0.2515
Category	0.8338	0.8094	0.6417
Name	0.8338	0.7667	0.3656
Phone	0.8554	0.8554	0.6929
Website	0.8323	0.8323	0.3483
Macro Avg	0.8370	0.8195	0.4600

My Approach

- Main goal: make ML more trustworthy, not just more complex
- I improved:
 - synthetic data realism using Yelp
 - size of the labeled benchmark: 200 -> 400
 - ML confidence quality and decision thresholds
 - the interaction between ML and rule-based methods
- Shift in strategy:
 - from ML vs rules
 - to ML selectively combined with rules

Why Yelp

- Large-scale business records with the same fields as the task
- Real labels were too limited for ML-only training
- Yelp made it possible to generate many realistic training pairs cheaply

Why More Labels

- The task has ambiguous edge cases
- More labels improve evaluation stability and confidence in conclusions

How I Improved ML

- Per-attribute models instead of one global model
- Improved synthetic generation:
 - attribute-specific corruption
 - near-equal cases
 - both-noisy cases
 - labels based on quality, not confidence shortcuts
- Added calibration-aware confidence with Platt scaling
- Tuned decision thresholds for F1 instead of fixed 0.5

Why calibration and threshold tuning mattered

- Raw probabilities are often overconfident
- Calibration makes confidence better match actual correctness
- Different attributes need different cutoffs
- I optimized directly for validation F1

Milestone	Address	Category	Name	Phone	Website	ML
						Macro F1
Starting ML	0.2515	0.6417	0.3656	0.6929	0.3483	0.4600
F1	0.7001	0.8220	0.6110	0.8000	0.8220	0.7001

Hybrid Strategy

- Original Mayhem hybrid:
 - heuristic combination of rule signals
 - still fully rule-based
 - no learned routing between ML and rules
- My hybrid idea:
 - baseline is the safe default
 - ML only overrides when confidence is strong enough
 - routing policy differs by attribute
- I searched 541,696 policy configurations

Example decision

- Baseline says base
- ML says current
- calibrated ML confidence = 0.91
- name policy threshold = 0.85
- final choice = ML

If confidence were 0.68, the router would fall back to the baseline.

Best final policy

Findings, Limitations, and Company Relevance

Findings

- Strong baselines can hide fragile logic
- ML improved a lot with better realism, calibration, and thresholds
- Best system design was selective override, not full ML replacement
- Confidence quality mattered almost as much as classifier quality

Limitations

- Synthetic-to-real gap still exists
- Real labeled benchmark is still relatively small
- Learned routing is data-hungry
- Some report artifacts still use legacy wording

Why this matters to a company

- Do not over-trust a shortcut baseline just because it wins one benchmark
- Safer deployment pattern:
 - keep strong heuristics
 - use calibrated ML selectively